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## Experimental physics with a homemade cycling power meter: development and characterization

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### Abstract

We report on a comprehensive, student-oriented approach to developing a low-cost, high-performance cycling power meter that integrates knowledge from physics, mechanics, electronics, and computer science. The power meter measures the force applied by the cyclist to the pedal crank using strain gauges arranged in a Wheatstone bridge configuration. In addition, a gyroscope is used to measure pedaling cadence. Instantaneous power is analyzed using the phyphox smartphone application, and the average power is calculated and transmitted to a bike computer or smartwatch via the Cycling Power Service, a Bluetooth Low Energy profile designed for transmitting cycling power data. The averaging process can be customized by adjusting the number of pedal rotations and applying a moving average, allowing for tailored power feedback to optimize training sessions. The cycling power meter provides accurate and reliable measurements, comparable to those of a commercial sensor, with an estimated relative error lower than 2%. This experimental physics project can be performed by students at minimal cost by following the step-by-step guide provided in this paper. The process encourages hands-on learning and the application of theoretical concepts in multiple disciplines, including data acquisition, signal processing, and wireless communication.

## I. INTRODUCTION

Engaging students in applying fundamental concepts from physics, mechanics, electronics, and programming through experimental projects is essential for developing their scientific and technical skills. Through this hands-on approach, they not only deepen their understanding of these interdisciplinary principles but also gain insight into measurement precision and the limitations of the involved devices. Moreover, project-based learning fosters the application of the scientific method and experimental techniques, while enhancing teamwork and organizational skills.<sup>1</sup> Beyond its educational value, the experimental physics project presented in the paper demonstrates how to create an original, low-cost sensor with practical applications in everyday life. More broadly, this work showcases an innovative use of the Arduino in combination with Bluetooth transmission, illustrating how this widely used microcontroller can be leveraged for physics experiments and data acquisition.<sup>2,3</sup>

When riding a bike, cyclists can easily monitor various physical parameters such as position, elevation, time, distance, speed, cadence, and heart rate.<sup>4</sup> To make these measurements, they simply need to equip their bike and body with relatively inexpensive sensors (e.g., GPS, cadence sensor, heart rate monitor). Although these parameters are useful, they do not inform the cyclist of the effort or energy expended to propel the bike forward. This additional parameter is the power, defined as the rate at which the cyclist generates mechanical energy to propel the bike forward.

Power meters have played a transformative role in modern cycling. The first commercial systems, introduced in the late 1980s and 1990s, were costly and often inaccurate. Over the last decade, technological advances have made these devices more reliable, although still relatively expensive. Today, power meters are ubiquitous in professional cycling, where almost every athlete uses them in both training and competition.<sup>5</sup> During races, team staff can monitor riders' real-time power output from a follow car, enabling data-driven decisions and precise pacing strategies. As a result, tactics are no longer based solely on intuition or experience, but increasingly on quantitative optimization enabled by accurate power measurement.

Power is a crucial metric because it directly measures the effort of the cyclist, regardless of external factors such as wind or terrain. It enables precise training, pacing, and performance analysis. Knowing their power output, cyclists can optimize their training zones, track

progress, and race more efficiently. For example, from the power delivered during a training session, cyclists can calculate the Maximal Aerobic Power (MAP), which is the highest power output sustained while consuming the maximum amount of oxygen (corresponding to the well-known  $\text{VO}_{2\text{max}}$ ). MAP is typically measured during an incremental exercise test in which intensity increases until exhaustion and represents the limit of aerobic capacity, making it a key indicator of endurance performance. MAP is for cycling what Maximal Aerobic Speed (MAS) is for running, where MAS is the lowest speed at which a runner reaches  $\text{VO}_{2\text{max}}$ . While MAP measures power in watts on a bike, MAS measures speed (m/s) during running, both reflecting aerobic capacity in two different sports contexts. From power measurements, one can also define the Functional Threshold Power (FTP) as the maximum constant power a cyclist can sustain for approximately one hour. FTP reflects the balance between aerobic and anaerobic energies and is widely used to establish training zones and compare the potential of athletes (e.g., FTP/kg). It is also a strong predictor of performance in time trials and endurance events. Therefore, measuring power in cycling is essential for accurately estimating the performance of a cyclist. However, this requires purchasing a power meter, which typically costs €500–1000. Consequently, power meters are often inaccessible to amateur cyclists and only make sense when associated with expensive top-end bikes.

Many homemade cycling power meter projects can be found online.<sup>6–10</sup> However, the explanations are often insufficient to successfully complete the work independently. In this paper, we provide a detailed step-by-step guide that clearly highlights the pitfalls to avoid. After reviewing the measurement principles and the instrumentation used, we present the measurement results and conclude with a comparison to a commercial cycling power meter. All programming codes used for this project will be made available in the Supplementary Material section. This project, which involves multidisciplinary knowledge in physics, mechanics, electronics, and computer science, can be carried out at low cost ( $\sim$  €50) by undergraduate or graduate students. It offers a practical application of theoretical concepts and encourages problem-solving skills through the development of a functional and innovative device.

## II. EDUCATIONAL CONTEXT: A CAPSTONE PROJECT FOR PHYSICS STUDENTS

The project was developed as a Capstone-style experience for senior undergraduate physics students specializing in instrumentation and applied physics. Designed as part of a hands-on learning approach, it engaged four students over five weeks of full-time work under minimal supervision, allowing them to integrate and apply the diverse knowledge they had acquired throughout their studies. The objective was to provide a rich, interdisciplinary learning experience that combines core competencies in mechanics, electronics, computer programming, signal processing, and wireless communication.

Students were tasked with developing a functional cycling power meter from scratch—conceiving the design, implementing sensor systems, acquiring and analyzing data, and transmitting the results via Bluetooth. The project required them to revisit and deepen their understanding of Newtonian mechanics (torque, rotational motion, power), strain gauge physics, analog-to-digital conversion, and filtering techniques. It also reinforced transferable skills such as teamwork, problem-solving, and scientific communication. Each student contributed to different facets of the work, including mechanical design, data acquisition, embedded programming, and signal analysis. The team met daily with an advisor and a technician, but retained a high degree of autonomy, encouraging ownership and independent problem-solving. The final assessment included both a written report and an oral presentation, emphasizing the importance of documenting and communicating scientific results.

This project serves as a model for advanced undergraduate instruction, offering a compelling framework for Capstone or senior thesis work in physics programs. Moreover, selected portions of the project — such as strain gauge calibration, signal processing of periodic data, or Bluetooth communication — can be adapted as classroom demonstrations or shorter lab modules. For instance, instructors can replicate simplified versions of the setup to illustrate real-time force measurement, the concept of mechanical power, or the application of strain gauges. It also serves as a strong foundation for interdisciplinary project-based learning, where students collaborate across physics, engineering, and computer science. By offering a concrete application with direct links to sports science and biomechanics, the project makes abstract physics concepts more tangible and engaging, thereby strengthening the pedagogical impact. In this way, the project not only enhances individual student competencies

but also contributes more broadly to physics education by bridging theory and real-world application.

### III. PRINCIPLE OF THE MEASUREMENT

The physics of cycling has been extensively studied, considering multiple factors influencing bicycle motion.<sup>11</sup> These include internal mechanical components such as the wheels, chain, front and rear derailleurs, and drivetrain efficiency, as well as external parameters such as road gradient, air resistance, rolling friction, and wind conditions. Researchers have analyzed the complex interactions between these elements to model cycling performance, optimize energy transfer, and improve efficiency.<sup>12–15</sup> In this study, we focus specifically on the power generated by the cyclist to propel the bicycle forward, quantifying the mechanical work exerted on the crank arms.

The power exerted by the cyclist is transmitted primarily through the pedals. From each pedal, this power is transferred to the crank arms and then to the wheels through the crankset, chain, and rear cassette attached to the rear wheel hub. To measure the power output, it is necessary to quantify the deformation caused by the force exerted by the cyclist on the pedals. In principle, this deformation can be measured in various locations on the bicycle, such as the crankset hub, the rear wheel hub, or on the crank arms themselves. In this study, we have chosen to measure the deformation on a single crank arm, specifically the left crank arm. In doing so, we only measure the power transmitted to the bike by the left leg of the cyclist. We then assume that the right leg contributes an equal amount of power, and estimate the total power output by doubling the measured value from the left crank arm. Typically, the discrepancies between the power contributions of the left and right legs are less than 2%. We will later ascertain whether this falls within the uncertainty range of the total power measurement.

Here is the mathematical formulation of power calculation. The deformation of the crank arm is measured and calibrated to obtain an equivalent mass  $M$  (kg) defined as  $F \equiv Mg$ , where  $F$  is the force applied by the cyclist's foot on the pedal and  $g = 9.81 \text{ m/s}^2$  is the acceleration due to gravity [Fig. 1(a)]. Using Abaqus software for finite element analysis and computer-aided engineering<sup>16</sup>, we simulated the deformation of an aluminum alloy crank under a 10 kg equivalent mass applied at the pedal thread. The simulation shows

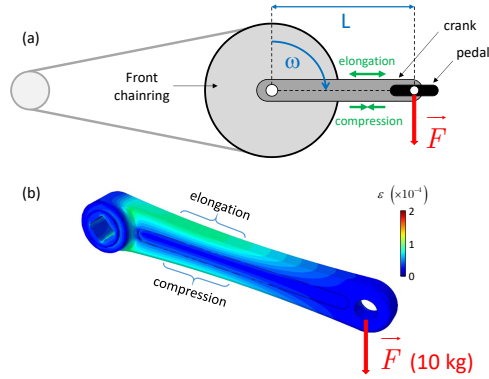


FIG. 1. (Color online) (a) Principle of the cycling power measurement. (b) Simulation of the aluminum alloy crank strain with Abaqus software. Color scale indicates the absolute value of the strain  $\epsilon = \Delta L/L$ .

the absolute value of the deformation, or strain, defined as  $\epsilon = \Delta L/L$ , where  $\Delta L$  is the crank elongation/compression at the position of the gauges [Fig. 1(b)]. The simulation illustrates the clear elongation and compression of the crank in the region closer to the rotation axis, which are both indicated by a green color since the simulation provides the absolute value of the crank strain under the action of the external force  $F$ . From this simulation, we conclude that the strain gauges should be fixed between the rotation axis and the middle of the crank, in order to obtain measurable signals from these sensors. It is important to note that the simulation is valid for the crank arm of our bike. The result may vary slightly if the material or shape of the crank arm is different, as the deformation is directly influenced by the Young modulus of the material.

As defined in Sec. I, power is the rate of energy transfer over time. In this context, the work done by the cyclist corresponds to the force  $F \equiv Mg$  exerted on the crank arm. This force produces a torque  $\Gamma = MgL$ , where  $L$  (m) is the crank length. The output power is then calculated as the product of this torque and the angular velocity  $\omega$  (rad/s) of the pedal stroke. Since the measurement is performed on the left crank arm, the total power output is obtained by assuming symmetry between the left and right legs. The total power  $P$  (W) is then :

$$P = 2MgL\omega. \quad (1)$$

This formulation provides a straightforward method to calculate the total power output of the cyclist, given the measured deformation of the left crank arm. To estimate the order of magnitude of the total power output, consider a left foot pressing on the pedal with an equivalent mass of  $M = 10$  kg, a crank arm length  $L = 0.175$  m, and a pedaling cadence  $PC$  (in units of rev/min, which is abbreviated as RPM henceforth) of 60 RPM corresponding to an angular velocity  $\omega = 2\pi \times PC/60 = 2\pi$  rad/s. Then, the total power is 215 W. Further analysis will assess the accuracy and potential uncertainties associated with this approach.

#### IV. MATERIAL

The power meter consists of four strain gauges, a 24-bit analog-to-digital converter (ADC) HX711 mounted on a printed circuit board and a Bluetooth-enabled Arduino Nano microcontroller (see Table I for more details on model numbers and distributors of the components). All components were purchased online, with the total system cost of less than €50. Figure 2 illustrates the arrangement between all these components, along with the final Bluetooth transmission from the Arduino Nano to the phyphox application or a bike computer.

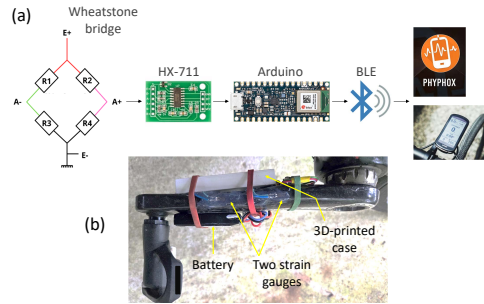


FIG. 2. (Color online) (a) Main elements of the homemade power meter. (b) The power meter mounted on the crank, along with the battery and the 3D-printed case including the HX711 and the Arduino. The two strain gauges on top side of crank are shown, while the two strain gauges on the bottom side of the crank are not visible.

The power meter uses four identical strain gauges ( $R_1$  to  $R_4$ ), each with a resistance of  $120 \Omega$ , configured in a full Wheatstone bridge to maximize sensitivity and temperature compensation (see Fig. 3). With this full-bridge configuration, the deformation of the crank arm



TABLE I. List of material

| Name            | Model                                                                                          | Price (€) |
|-----------------|------------------------------------------------------------------------------------------------|-----------|
| 4 strain gauges | Youmile BF120-3AA, resistor $120 \Omega$ , size $8.5 \times 4$ mm                              | 5         |
| 24-bit ADC      | Haljia HX711 module, size $34 \times 21 \times 6$ mm                                           | 2         |
| Microcontroller | Arduino Nano 33 BLE Sense Rev2 (incl. IMU BMI270), size $18 \times 42$ mm                      | 32        |
| Battery         | LiPo 2s, 7.4 V, 400 mAh, with a 5 V step-down regulator and 2S Battery Management System (BMS) | 10        |
| 3D printed case | homemade                                                                                       | 0         |

is accurately measured. This setup also compensates for lateral stresses, i.e., deformations of the crank arm in the direction perpendicular to the force applied by the foot on the pedal due to the Poisson effect. If the user wants to simplify the implementation of the sensor, it is possible to use only two strain gauges instead of four, but the sensor sensitivity will also be reduced by half. In this case, one gauge should be placed on top and the other on the bottom of the crank arm, with two additional  $120 \Omega$  resistors to complete the Wheatstone bridge. This optional setup still ensures compensation for temperature variations and the Poisson effect.

In a Wheatstone bridge powered by a voltage  $E_+$ , the output voltage  $\Delta V$  is measured between the midpoints of the two voltage dividers formed by the resistors. Figure 3 illustrates the wiring of the Wheatstone bridge on the crank. Based on the simulation shown in Fig. 1(b), we expect an increase in resistance for  $R_2$  and  $R_3$  (crank elongation) and a decrease for  $R_1$  and  $R_4$  (crank compression). Assuming that this change in resistance is equal to  $\Delta R$  for all resistors, the output voltage of the Wheatstone bridge is given by:

$$\Delta V = A_+ - A_- = E_+ \frac{R_2 R_3 - R_1 R_4}{(R_1 + R_2)(R_3 + R_4)} = E_+ \frac{\Delta R}{R}. \quad (2)$$

Since the gauge factor  $G_f$  is defined as  $G_f = \frac{\Delta R/R}{\epsilon} = \frac{\Delta R/R}{\Delta L/L}$ , we can write the relationship between the output voltage of the Wheatstone bridge and the crank deformation as:

$$\Delta V = E_+ G_f \frac{\Delta L}{L}, \quad (3)$$

with the value for  $G_f$  typically a little over 2 for common metallic foil gauges. This equation indicates that the bridge is at equilibrium in the absence of deformation and that the

measured voltage is directly proportional to the crank deformation. Moreover, by design, the Wheatstone bridge is not sensitive to any (perpendicular) force applied along the crank axis, as long as this force deforms each of the four resistors equally. Gluing the four strain

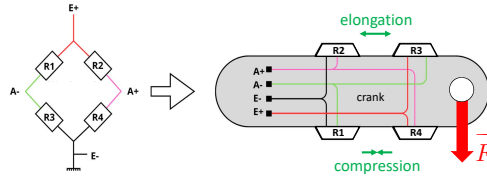


FIG. 3. (Color online) Wiring of the Wheatstone bridge on the crank. Follow color coding for appropriate wiring.

gauges does not present any particular difficulty. Many online tutorials provide guidance.<sup>17</sup> A special adhesive can be used, but cyanoacrylate glue is suitable. It is important to verify the correct wiring of the gauges (see the colored wires in Fig. 3) using an ohmmeter. Verify that the bridge impedance is  $120\ \Omega$   $((120 + 120)/(120 + 120))$  on the diagonals and  $90\ \Omega$   $(120/(120 + 120 + 120))$  on the four sides. Once the gauges are in place, it is recommended to cover them and the connecting wires with a protective coating, such as silicone, to prevent detachment and provide protection against external elements.

The HX711 ADC is used to amplify and digitize small voltage variations from the Wheatstone bridge. It features a programmable gain amplifier; we used the highest selectable gain of 128, making it suitable for measuring low-voltage differential input signals. The HX711 provides a 24-bit ADC for full-scale signal conversion, enabling precise force measurements. It is powered by the Arduino at 3.3 V. Figure 4 illustrates the wiring between the Wheatstone bridge, the HX711, and the Arduino microcontroller. By default, the HX711 operates at a sampling rate of 10 Hz, which is insufficient for precise measurement. This rate has been increased to 80 Hz by modifying a pin in the module, as indicated by the yellow wire in Fig. 4, which connects the RATE pin to the DVDD pin. This sampling rate is well-suited to measure the crank deformation for a usual cycling cadence (e.g., less than  $\text{RPM} = 140$  rev/min). The AVDD pin of the HX711 provides a regulated voltage. Therefore, no external power supply should be connected to the  $E_+$  output of the HX711 to power the Wheatstone bridge. The HX711 is powered through the DVDD pin, which is supplied with 3.3 V from the Arduino. Connecting the  $E_-$  pin of the HX711 to the circuit's common

ground is unnecessary, since the AGND pin is already internally connected to ground.

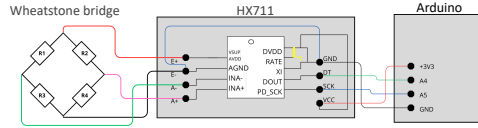


FIG. 4. (Color online) Global wiring between the Wheatstone bridge, the HX711 and the Arduino.

The core of the power meter is the Arduino Nano 33 BLE Sense Rev2 microcontroller, which features a 6-axis integrated inertial measurement unit (IMU), the BMI270, capable of measuring the angular velocity along three axes. Using Arduino IDE software,<sup>18</sup> we developed a program to calculate power and cadence (code available in the Supplementary Material). The Arduino operates at 3.3 V and communicates wirelessly via Bluetooth Low Energy (BLE) to transmit power and cadence data. BLE has been preferred to the other ANT+ transmission protocol, a proprietary wireless communication protocol developed by Garmin that is commonly used in sports and fitness applications, since it is supposed to be discontinued in June 2025. The data transmission rate is 80 Hz, limited by the sampling frequency of the HX711, ensuring real-time monitoring on compatible devices such as cycling computers or smartwatches.

The system is powered by a rechargeable battery. To estimate the required capacity, the total current consumption of the system is calculated as 44 mA, broken down as follows: HX711 (1.5 mA), Arduino (15 mA in active mode), and Wheatstone bridge (27.5 mA for four 120  $\Omega$  resistors powered at 3.3 V). A 2S lithium polymer (LiPo) battery with a nominal voltage of 7.4 V and a capacity of 400 mAh was selected to ensure sufficient runtime while maintaining a compact form factor. The LiPo 2S battery is ideal because it is compact, affordable, and offers great autonomy. However, it must not discharge below approximately 6 V, as this could cause irreversible damage. To prevent this, it should be paired with a 5 V step-down regulator (e.g., AMS1117 5V or MP2315, requiring an input voltage between 6.5 and 12 V) to safely power the Arduino. To further protect the battery, a 2S Battery Management System (BMS) should be used. This circuit prevents over-discharge by cutting off the power supply if the voltage drops below a safe threshold, typically around 6 V, thereby extending battery lifespan and ensuring safe operation.

A custom 3D-printed enclosure was designed to securely house the HX711, Arduino and

battery, protecting the electronics from environmental factors and mechanical vibrations encountered during cycling.

We used the phyphox smartphone application,<sup>19</sup> developed by RWTH Aachen University, to collect data transmitted by the Arduino via the BLE protocol.<sup>20</sup> This application uses the sensors of a smartphone to produce data and/or graphs in real-time.<sup>21</sup> This functionality requires the installation of the phyphoxBle.h Arduino library in the Arduino IDE. A tutorial is available online in French.<sup>22</sup> Phyphox allows users to create their own experiments using a visual experiment webeditor,<sup>23</sup> which will generate a simple file that defines your experiment including data analysis. This file, available in the Supplementary Material, allows one to plot and record the cadence and power as a function of time at 80 Hz, allowing data analysis and optimization of the final algorithm for mean cadence and power calculations. The file must be loaded in the phyphox webeditor in order to generate a QR code. Then, using the smartphone and the phyphox application, the QR code must be flashed to generate the experiment. After this analysis, we implemented a second Arduino program to compute these parameters and transmit them via BLE to a bike computer or a smartwatch (e.g., Apple watch, Garmin, Coros, Polar). For this purpose, we used the Cycling Power Service (CPS),<sup>24,25</sup> a Bluetooth profile specifically designed to transmit cycling power data (code available in the Supplementary Material). Using this protocol, the cyclist can connect to the Arduino by searching for a Power service on their bike computer or smartwatch. Special attention was paid to the output format of the cadence, which must be calculated using the cumulative crank revolutions and the timestamp of the last crank revolution, measured in units of 1/1024 seconds. A key advantage of using the CPS protocol is that training sessions are automatically recorded and analyzed using commercial tools provided by the bike computer or smartwatch manufacturer.

## V. RESULTS

In this section, we first verify the accurate measurement of the cadence provided by the Arduino gyroscope. Next, we calibrate the force measurements obtained from the HX711 ADC and perform real-time measurements of pedaling cadence and power using the phyphox application. Finally, we will compare the transmitted data under real training conditions with a commercial power meter.

### 1. Cadence measurement using the Arduino gyroscope

This first experiment consists in positioning the Arduino on a rotating plate with known angular velocity. For that purpose, we used a DDR25 Thorlabs motorized rotation mount piloted by a servo driver, which gives an accurate measurement of the rotation speed. If this equipment is not available, it is possible to measure the rotation speed of the stage using a homemade optical gate using a light source, a photodiode, and an oscilloscope. Another simple solution consists of using a magnet, the magnetometer of a smartphone, and a Fourier transform to analyze the magnetic field variations over time. Figure 5 shows the measurement obtained with the Arduino (in RPM) as a function of the RPM fixed by the driver of the motorized stage. The linear fit of the experimental data is  $y = (1.012 \pm 0.002)x - (0.5 \pm 0.2)$  (RPM). The inset illustrates the fluctuations of the Arduino measurement over time when the rotation is fixed to 100 RPM. The standard deviation of the data is 0.54 RPM. We then conclude that the Arduino gyroscope can measure the rotation speed with a relative error better than 0.6% around 100 RPM. We verified that the measurement is not affected when the Arduino is tilted a few degrees relative to the plane of rotation. This indicates that the measurement will not be overly sensitive to the positioning of the Arduino on the bike crank.

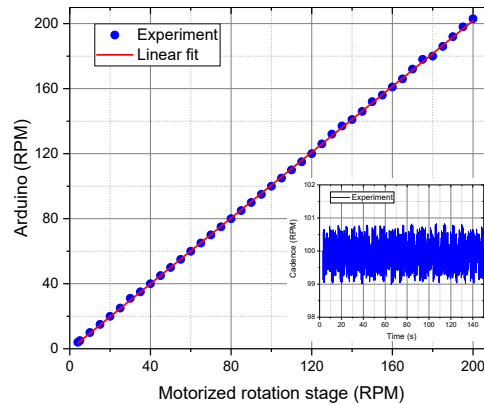


FIG. 5. (Color online) Cadence measured by the Arduino as a function of reference cadence imposed by a motorized rotation stage. Inset: evolution of the cadence measured by the Arduino over time.

## 2. Calibration of the HX711 ADC

The second measurement is essential and must be performed for each sensor since it depends on crank material and shape. This is undoubtedly a significant limitation if the sensor we are developing were to be installed by a general user. This measurement concerns the calibration to convert the HX711 ADC output into force values, or equivalent masses applied to the crank. To achieve this calibration, with the crank in a horizontal position and after zeroing with no load (tare measurement), we attached calibrated weights from 0 to 22 kg and recorded the values measured by the HX711 ADC. Figure 6 shows that the crank deformation remains perfectly elastic under these conditions, with a linear fit of the measurements corresponding to  $y = (22370 \pm 90)x + (200 \pm 100)$  (a.u.). Later, we will verify with pedaling measurements on the bike whether the 0 to 22 kg range is sufficient. In conclusion, the calibration coefficient for the crank in this experiment is 22370 with a relative error of 0.4%. This calibration will be applied hereafter for power measurement. Additional endurance test measurements, such as the evolution of the ADC value over time for a fixed load, are not useful since they do not reflect realistic cycling conditions.

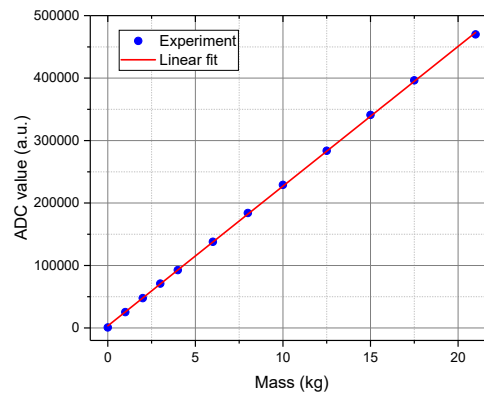


FIG. 6. (Color online) Calibration of the crank deformation. Evolution of the HX711 ADC value as a function of the mass suspended on the crank.

### 3. Real-time measurement of the power using *phyphox*

After the calibration of the crank arm, we are able to calculate the power. Before sending this information to the cyclist, it is necessary to analyze the evolution of cadence and power over time during crank rotation. This is achieved at 80 Hz via BLE transmission to the *phyphox* application, as explained in Sec. IV. Before powering the Arduino, the pedal must be placed in a vertical position so that the automatic tare calibration performed by the Arduino code is performed without any stress on the crank arm. Thus, remember that one should calibrate with a horizontal pedal but tare with a vertical one.

The experiment consists of a 20 s pedaling session at a cadence of approximately 60 RPM. Figure 7(a) illustrates the evolution of the pedaling cadence over time. As previously described, this measurement is accurate to within  $\pm 0.5$  RPM, which is confirmed again here by the standard deviation analysis. With a relative error of 0.6% for the cadence around 100 RPM, we decided not to reduce this uncertainty by using averaging techniques to avoid slowing down of the cadence and power computation algorithm. Figure 7(b) shows the evolution of the force exerted by the cyclist on the pedal equipped with the sensor. This measurement is obtained using the calibration performed earlier (Sec. V 2). For simplicity, we plotted the evolution of the equivalent mass exerted on the pedal over time. Although this force measurement is not required for power calculation, the graph is highly useful as it allows us to verify whether the calibration was performed over a sufficient force range. We observe that the maximum mass applied on the pedal is approximately 15 kg. Even if the cyclist pedals with twice this power, we conclude that the calibration performed between 0 and 22 kg is appropriate.

Figure 7(b) also reveals an unusual behavior during the first 10 pedal strokes (between 0 and 10 s). The measured mass on the left pedal becomes periodically negative, reaching a minimum of approximately  $-7.5$  kg when the pedal is near the rear wheel, corresponding to the phase in which the cyclist no longer pushes on this pedal to generate forward motion. This behavior is not unexpected. Since the sensor functions like a scale, it records the deformation of the crank arm caused by the vertical component of the applied force. When the crank is in the forward position, the weight of the leg increases the deformation in one direction; conversely, when the crank is in the rearward position, the same weight induces the opposite deformation. The force is therefore recorded as negative when the pedal is

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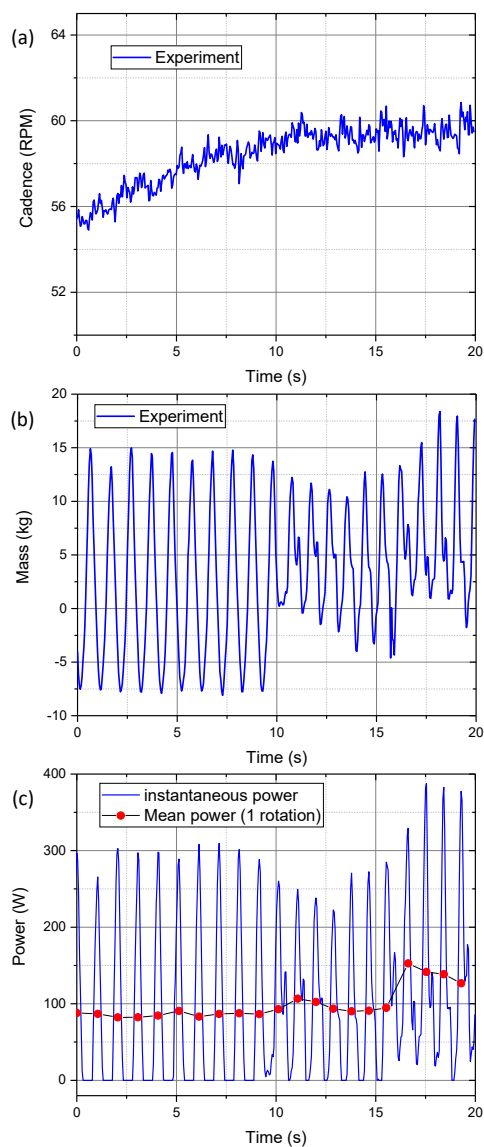


FIG. 7. (Color online) Real-time measurement using phyphox. (a) Cadence. (b) Force applied on the pedal, represented by the corresponding effective mass. (c) Power developed by the cyclist to propel the bike forward. Red circles: mean power per pedal stroke.



in the rear. This interpretation was confirmed through a control experiment in which the crank rotated freely with no foot on the pedal. In this case, the measured force was positive in the forward position and negative in the rear, assuming an initial tare when the crank is vertical. The presence of a negative force even without any active input shows that this effect is primarily geometric and due to the crank orientation and the gravitational force acting on the pedal and leg, rather than from any dynamic elastic overshoot. Between 10 and 20 s, the cyclist voluntarily applied force during the upstroke by pulling up on the pedal, thanks to the use of pedals with cleats that secure the shoes. During this interval, the negative force disappeared from the graph, indicating that the crank no longer experienced the opposite deformation. This demonstrates that the system is capable of capturing subtle changes in pedaling technique, and highlights the potential of the sensor setup for detailed biomechanical analysis.

When calculating the instantaneous power using Eq. (1), we set the force to zero whenever it is negative, as these phases do not contribute to forward propulsion. Although this is a simplified approach, it ensures that only power-generating segments are considered. In the future, a more advanced algorithm could take into account negative or pulling forces, when appropriate, particularly for experienced cyclists who employ more complete pedaling techniques. Figure 7(c) shows the evolution of power over time. It varies between 0 and nearly 400 W, which is compatible with usual power values for a standard effort on a bike. During the first 10 s, when the cyclist does not pull on the pedals in the backward phase, the power logically drops to zero during part of the pedaling cycle, reducing pedal efficiency. However, as seen after 10 s, the graph clearly shows that efficiency improves as soon as the cyclist starts pulling on the pedal during its return phase. This effect becomes more evident when calculating the average power per pedal stroke, obtained by summing the total power over one full crank rotation and dividing by the rotation duration (red circles on the curve). Between 15 and 20 s, we observe a clear increase in the cyclist's power output.

We can also estimate the accuracy of the power measurement. From Eq. (1), around 100 RPM, the relative power error of the sensor is  $\Delta P/P = \Delta M/M + \Delta(\text{RPM})/\text{RPM} = 1.0\%$ . It arises from errors in cadence (0.6%, as seen in Sec. V 1), and force applied to the pedal (0.4%, as seen in Sec. V 2). We also have to take into account the sampling rate of 80 Hz, even if this rate is well adapted to a cadence up to 160 RPM, which is already very fast. We conclude that the relative power error is less than 2%. Assuming that the average

power remains constant between 0 and 10 s in Fig. 7(c), we obtain  $P = (85.8 \pm 2.7)$  W over this period, corresponding to a relative error of 3.0%. This is slightly above the expected 2%, but we have to take into account the intrinsic pedaling variations during the experiment, as the cyclist cannot maintain a perfectly constant power output over 10 s.

#### 4. Final data transmission to the bike monitor

We finally used the second Arduino program, available in the Supplementary Material, to transmit power and cadence to the cyclist through BLE. As explained in Sec. IV, this is done using the CPS profile, and training sessions can be automatically recorded and analyzed using commercial tools provided by a bike computer or smartwatch. We even succeeded in connecting the homemade power meter to an indoor bike application such as Zwift to participate in online races. As seen in Sec. V 3, real-time power is not relevant for the cyclist. Therefore, the code calculates the mean power and cadence on an adjustable number of pedal rotation. For reactivity, in this section, results correspond to a single pedal rotation. In order to reduce fluctuations, we also implemented a moving average of the data as follows. At each new power and cadence measurement, the oldest value is discarded, and the newest value is included in the calculation, ensuring a continuously updated average. In this section, we chose to apply the moving average over three values to reduce fluctuations while maintaining responsiveness.

For this final experiment, our aim was to compare the measurements of our homemade power meter with those obtained from a commercial sensor. We mounted the bike's rear cassette on a Wahoo smart trainer (model KICKR Core,  $\sim$ €500), which is equipped with a Bluetooth transmitter that transmits cadence and power data to a GPS receiver (Garmin Edge 540). According to the manufacturer, this sensor provides a measurement accuracy of  $\pm 2\%$  for both cadence and power. We pedaled for 20 minutes and simultaneously recorded the cadence and power data transmitted by both sensors.

Figure 8(a) illustrates the evolution of cadence over time. The similarity between the measurements from our sensor and the commercial power meter is striking, even when the cyclist suddenly increases the cadence, as observed at times 940 and 1070 s. The cadence measurement accuracy of our sensor was previously estimated at approximately  $\pm 0.6\%$ , which is even better than that of the commercial sensor. Additionally, we note that the

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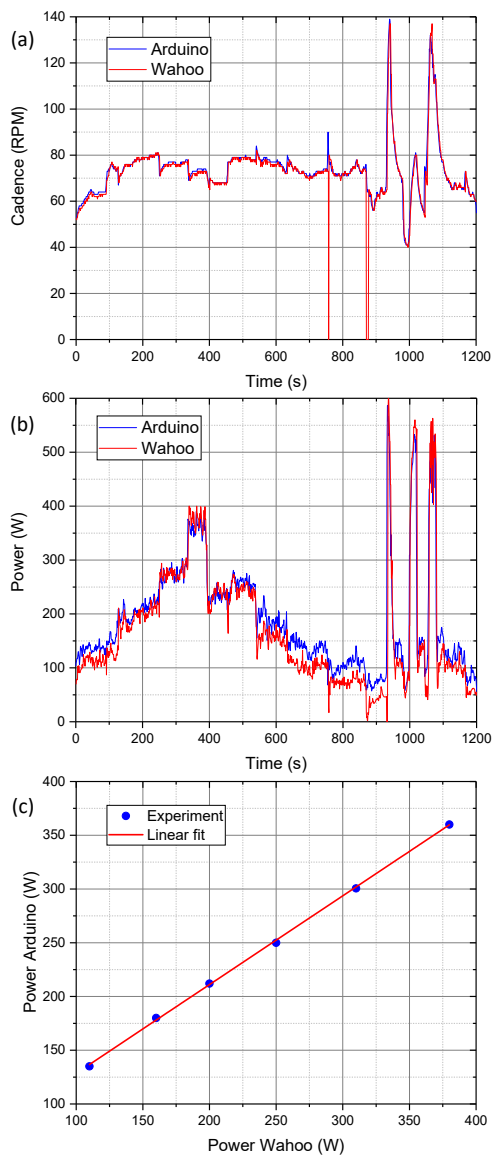


FIG. 8. (Color online) Comparison with a commercial cycling power meter. (a) Cadence. (b) Power. (c) Difference between the homemade power meter and the commercial one.

commercial sensor occasionally loses cadence data (once at times 760 s and twice around 870 s), whereas our sensor remains perfectly stable throughout the entire recording sequence. Figure 8(b) shows the evolution of power over time for both sensors. The similarity between

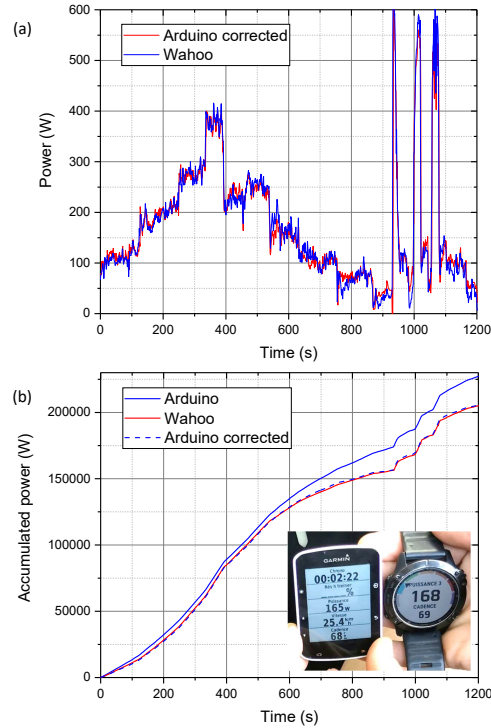


FIG. 9. (Color online) Comparison with a commercial power meter after numeric correction of the homemade power meter. (a) Power. (b) Accumulated power during the training session. Inset: picture at a given time of the bike computer connected to the commercial power meter (left) and the smartwatch connected to the homemade power meter (right). *Puissance* means *power* in French.

the two measurements remains very strong, even during the two sprint phases, at 940 and 1070 s. This result is crucial, as it validates the ability of our homemade sensor to accurately measure and transmit power via Bluetooth under real training conditions. However, upon closer inspection of the two curves, we observe slight discrepancies between the measurements from the two sensors. This is not surprising, as power measurement is performed

in entirely different ways. Our sensor determines power based on the deformation of the crank arm, whereas the commercial sensor measures it at the rear cassette. Between these two measurement points, several mechanical factors come into play, such as chain transmission losses. Additionally, the deformation measured by the commercial sensor occurs at every pedal stroke (both left and right), meaning that its power output does not need to be doubled, unlike our sensor. Given these fundamental differences in measurement methods, the results are expected to be not exactly the same. We have plotted these discrepancies in Fig. 8(c) for six power values ranging from 110 W to 380 W. The data points are well represented by the linear fit  $P_{\text{Arduino}} = (0.826 \pm 0.009)P_{\text{Wahoo}} + (46 \pm 2) \text{ (W)}$ , where  $P_{\text{Arduino}}$  is the power measured by our sensor and  $P_{\text{Wahoo}}$  that of the commercial one. With this result, we can apply a correction to our sensor to better align with the commercial sensor if we consider that it can serve as a reference. We also conclude that the chain transmission losses, while typically small and consistent, leads to a systematic and approximately linear discrepancy between the two measurements, as the drivetrain efficiency remains relatively constant during the test conditions.

The result of this correction is presented in Fig. 9(a), where we have plotted  $P_{\text{Arduino-corrected}} = (P_{\text{Arduino}} - 46)/0.826$  over time. Now, the two power curves are almost identical, demonstrating that the correction is highly effective within the considered power range. This result also illustrates that both sensors have the same accuracy, estimated to be better than 2%. To further confirm the effectiveness of the correction, we calculated the cumulative power over time generated by the cyclist [Fig. 9(a)]. Before correction, the cumulative power measured by our sensor (solid blue line) overestimates the power compared to the commercial sensor (solid red line). However, after applying the correction, the cumulative power from our sensor (dotted blue line) aligns almost perfectly with that of the commercial sensor.

## VI. CONCLUSION

In this study, we have successfully developed and characterized a low-cost homemade cycling power meter, integrating concepts of physics, mechanics, electronics, and computer science. The system is based on four strain gauges and a HX711 ADC to measure the deformation of the crank arm, combined with an IMU gyroscope to determine the pedaling cadence. Data are processed by an Arduino microcontroller and transmitted via Bluetooth,

enabling real-time monitoring on a bike computer or smartwatch.

Our results demonstrate that the power meter provides accurate and reliable measurements, comparable to those of a commercial sensor, with an estimated relative error less than 2%. The validation experiments showed strong agreement between the two systems, even during sprint phases. In addition, a correction factor was applied to further align the data of our sensor with those of a commercial power meter, highlighting its robustness for practical cycling applications.

To further improve the system, a key perspective is to simplify the design by integrating all components onto a custom Printed Circuit Board (PCB), eliminating unnecessary wiring and manual soldering. This would improve the durability of the device and the ease of assembly, making it more accessible to students and cyclists. Automation of the calibration procedure is currently in progress using a Visual Basic for Applications (VBA) script. Lastly, to reduce costs while maintaining performance, the Arduino Nano 33 BLE Sense could be replaced by a Seeed Studio XIAO nRF52840 Sense microcontroller, which also features Bluetooth transmission and an integrated IMU. These optimizations would make the power meter even more compact, cost-effective, and user-friendly, paving the way for broader adoption and potential further developments.

Beyond its technical implementation, this project exemplifies a Capstone-style educational experience that integrates core areas of physics with hands-on engineering practice. It not only helps students develop interdisciplinary skills but also serves as a model for advanced undergraduate instruction. In addition, key elements of the project — such as strain gauge calibration, data acquisition, and signal processing — can be adapted into classroom demonstrations or modular lab exercises, providing versatile teaching tools within the physics curriculum.

#### SUPPLEMENTARY MATERIAL

Please click on this link to access the supplementary material, which includes three Arduino codes (crank calibration, real-time measurement of the power using phyphox, final data transmission to the bike monitor) and one phyphox code. Print readers can see the supplementary material at [DOI to be inserted by AIPP].

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## AUTHOR DECLARATIONS SECTION

The authors have no conflicts to disclose. R.B., M.F., V.G., and Q.T. are students and carried out the experiments under the supervision of P.M. and E.A. The paper was written by E.A. and approved by all authors.

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